Problem Set 6

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2026-02-25

Problem set 6: Model Selection

To demonstrate the application of subset selection methods

Consider the “Hitters” data. For loading the data you need to install ISLR library. We wish to predict a baseball player’s salary on the basis of various predictors. So first we need to choose an optimal model.

```
rm(list=ls())
set.seed(123)
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.5.2

library(leaps)

## Warning: package 'leaps' was built under R version 4.5.3

str(Hitters)

## 'data.frame':    322 obs. of  20 variables:
##  $ AtBat    : int  293 315 479 496 321 594 185 298 323 401 ...
##  $ Hits     : int  66 81 130 141 87 169 37 73 81 92 ...
##  $ HmRun    : int  1 7 18 20 10 4 1 0 6 17 ...
##  $ Runs     : int  30 24 66 65 39 74 23 24 26 49 ...
##  $ RBI      : int  29 38 72 78 42 51 8 24 32 66 ...
##  $ Walks    : int  14 39 76 37 30 35 21 7 8 65 ...
##  $ Years    : int  1 14 3 11 2 11 2 3 2 13 ...
##  $ CAtBat   : int  293 3449 1624 5628 396 4408 214 509 341 5206 ...
##  $ CHits    : int  66 835 457 1575 101 1133 42 108 86 1332 ...
##  $ CHmRun   : int  1 69 63 225 12 19 1 0 6 253 ...
##  $ CRuns    : int  30 321 224 828 48 501 30 41 32 784 ...
##  $ CRBI     : int  29 414 266 838 46 336 9 37 34 890 ...
##  $ CWalks   : int  14 375 263 354 33 194 24 12 8 866 ...
##  $ League   : Factor w/ 2 levels "A","N": 1 2 1 2 2 1 2 1 2 1 ...
##  $ Division : Factor w/ 2 levels "E","W": 1 2 2 1 1 2 1 2 2 1 ...
##  $ PutOuts  : int  446 632 880 200 805 282 76 121 143 0 ...
##  $ Assists  : int  33 43 82 11 40 421 127 283 290 0 ...
##  $ Errors   : int  20 10 14 3 4 25 7 9 19 0 ...
##  $ Salary   : num  NA 475 480 500 91.5 750 70 100 75 1100 ...
##  $ NewLeague: Factor w/ 2 levels "A","N": 1 2 1 2 2 1 1 1 2 1 ...
```

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```r
hitter_clean = na.omit(Hitters)
head(hitter_clean)
```

```
##                    AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits
CHmRun
## -Alan Ashby         315   81     7   24  38    39    14   3449   835
69
## -Alvin Davis        479  130    18   66  72    76     3   1624   457
63
## -Andre Dawson       496  141    20   65  78    37    11   5628  1575
225
## -Andres Galarraga   321   87    10   39  42    30     2    396   101
12
## -Alfredo Griffin    594  169     4   74  51    35    11   4408  1133
19
## -Al Newman          185   37     1   23   8    21     2    214    42
1
##                    CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Alan Ashby         321  414    375      N        W     632      43     10
## -Alvin Davis        224  266    263      A        W     880      82     14
## -Andre Dawson       828  838    354      N        E     200      11      3
## -Andres Galarraga    48   46     33      N        E     805      40      4
## -Alfredo Griffin    501  336    194      A        W     282     421     25
## -Al Newman           30    9     24      N        E      76     127      7
##                    Salary NewLeague
## -Alan Ashby         475.0         N
## -Alvin Davis        480.0         A
## -Andre Dawson       500.0         N
## -Andres Galarraga    91.5         N
## -Alfredo Griffin    750.0         A
## -Al Newman           70.0         A
```

2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```r
n = nrow(hitter_clean)
train = sample(1:n, floor(0.8*n))
train_data = hitter_clean[train, ]
test_data = hitter_clean[-train, ]

head(train_data)
```

```
##                  AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
CRuns
## -Marvell Wynne   288   76     7   34  37    15     4   1644   408     16
198
## -Bill Almon      196   43     7   29  27    30    13   3231   825     36
376
## -Rick Cerone     216   56     4   22  18    15    12   2796   665     43
266
## -Jack Howell     151   41     4   26  21    19     2    288    68      9
```

```
45
## -Spike Owen      528  122     1   67  45    51     4   1716   403     12
211
## -Tom Paciorek    213   61     4   17  22     3    17   4061  1145     83
488
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Marvell Wynne  120    113      N        W     203       3      3    240
## -Bill Almon     290    238      N        E      80      45      8    240
## -Rick Cerone    304    198      A        E     391      44      4    250
## -Jack Howell     39     35      A        W      28      56      2     95
## -Spike Owen     146    155      A        W     209     372     17    350
## -Tom Paciorek   491    244      A        W     178      45      4    235
##                NewLeague
## -Marvell Wynne         N
## -Bill Almon            N
## -Rick Cerone           A
## -Jack Howell           A
## -Spike Owen            A
## -Tom Paciorek          A
```

```r
head(test_data)
```

```
##                AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
CRuns
## -Alvin Davis     479  130    18   66  72    76     3   1624   457     63
224
## -Andre Dawson    496  141    20   65  78    37    11   5628  1575    225
828
## -Andy VanSlyke   418  113    13   48  61    47     4   1512   392     41
205
## -Buddy Bell      568  158    20   89  75    73    15   8068  2273    177
1045
## -Barry Bonds     413   92    16   72  48    65     1    413    92     16
72
## -Bobby Bonilla   426  109     3   55  43    62     1    426   109      3
55
##                CRBI CWalks League Division PutOuts Assists Errors Salary
## -Alvin Davis    266    263      A        W     880      82     14    480
## -Andre Dawson   838    354      N        E     200      11      3    500
## -Andy VanSlyke  204    203      N        E     211      11      7    550
## -Buddy Bell     993    732      N        W     105     290     10    775
## -Barry Bonds     48     65      N        E     280       9      5    100
## -Bobby Bonilla   43     62      A        W     361      22      2    115
##                NewLeague
## -Alvin Davis           A
## -Andre Dawson          N
## -Andy VanSlyke         N
## -Buddy Bell            N
## -Barry Bonds           N
## -Bobby Bonilla         N
```

```r
attach(train_data)
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model.Choose the best model according to adjusted R^2, BIC, AIC and C_p values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

```r
reg1 = regsubsets(Salary ~ ., data = train_data, nvmax = 19)
reg1_sum = summary(reg1)
```

Best Subset selection according Aadjusted R^2.

```r
c_adjr2 = coef(reg1,which.max(reg1_sum$adjr2))
c_adjr2

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CRuns
##  154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154
1.7712871
##         CRBI       CWalks    DivisionW      PutOuts      Assists
Errors
##    1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -
6.4726568
##   NewLeagueN
##   75.8378982

predictors = c(names(c_adjr2)[-c(1,9,13)],"Division","NewLeague")
predictors

##  [1] "AtBat"     "Hits"      "Walks"     "CAtBat"    "CRuns"     "CRBI"
##  [7] "CWalks"    "PutOuts"   "Assists"   "Errors"    "Division"
"NewLeague"

formula = reformulate(termlabels = predictors, response = "Salary")
formula

## Salary ~ AtBat + Hits + Walks + CAtBat + CRuns + CRBI + CWalks +
##     PutOuts + Assists + Errors + Division + NewLeague

m1 = lm(formula, train_data)

y_h1 = predict(m1)
y_test1 = predict(m1,newdata = test_data)

train_mse1 = mean((train_data$Salary - y_h1)^2)
train_mse1

## [1] 88935.8
```

```
test_mse1 = mean((test_data$Salary - y_test1)^2)
test_mse1

## [1] 130066.9
```

Best Subset selection according BIC

```
c_bic = coef(reg1,which.min(reg1_sum$bic))
c_bic

##  (Intercept)        Walks       CAtBat        CHits       CHmRun
DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012

predictors = c(names(c_bic)[-c(1,6)],"Division")
predictors

## [1] "Walks"    "CAtBat"   "CHits"    "CHmRun"   "Division"

formula = reformulate(termlabels = predictors, response = "Salary")
formula

## Salary ~ Walks + CAtBat + CHits + CHmRun + Division

m2 = lm(formula, train_data)

y_h2 = predict(m2)
y_test2 = predict(m2,newdata = test_data)

train_mse2 = mean((train_data$Salary - y_h2)^2)
train_mse2

## [1] 97026.43

test_mse2 = mean((test_data$Salary - y_test2)^2)
test_mse2

## [1] 153090.5
```

Best Subset selection according Mallow's C_p measure:

```
c_cp = coef(reg1,which.min(reg1_sum$cp))
c_cp

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CRuns
##  134.1593585   -1.9328699    5.8271814    6.5037245   -0.2063323
1.7820180
##         CRBI       CWalks    DivisionW      PutOuts      Assists
NewLeagueN
##    1.1120229   -0.8879140 -121.2135304    0.1925628    0.3316973
71.1433129
```

```
predictors = c(names(c_cp)[-c(1,9,12)],"Division","NewLeague")
predictors
```

```
##  [1] "AtBat"     "Hits"      "Walks"     "CAtBat"    "CRuns"     "CRBI"
##  [7] "CWalks"    "PutOuts"   "Assists"   "Division"  "NewLeague"
```

```
formula = reformulate(termlabels = predictors, response = "Salary")
formula
```

```
## Salary ~ AtBat + Hits + Walks + CAtBat + CRuns + CRBI + CWalks +
##     PutOuts + Assists + Division + NewLeague
```

```
m3 = lm(formula, train_data)

y_h3 = predict(m3)
y_test3 = predict(m3,newdata = test_data)

train_mse3 = mean((train_data$Salary - y_h3)^2)
train_mse3
```

```
## [1] 89790.63
```

```
test_mse3 = mean((test_data$Salary - y_test3)^2)
test_mse3
```

```
## [1] 125652.3
```

4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```
Measure = c("Adjusted R sq.","BIC","Mallow's Cp")
Train_MSE_1 = c(train_mse1,train_mse2,train_mse3)
Test_MSE_1 = c(test_mse1,test_mse2,test_mse3)

#Comparision
output = data.frame(Measure,Train_MSE_1,Test_MSE_1)
output
```

```
##          Measure Train_MSE_1 Test_MSE_1
## 1 Adjusted R sq.    88935.80   130066.9
## 2            BIC    97026.43   153090.5
## 3    Mallow's Cp    89790.63   125652.3
```

5. Repeat 3-4 using forward stepwise selection method.

FORWARD STEPWISE SELECTION

```
reg2 = regsubsets(Salary ~ ., data = train_data, method = "forward", nvmax =
19)
reg2_sum = summary(reg2)
```

Forward Subset selection according Aadjusted R^2.

```
c_adjr2 = coef(reg2,which.max(reg2_sum$adjr2))
c_adjr2

##  (Intercept)        AtBat         Hits        Walks        Years
CAtBat
##  207.8865937   -1.6603271    4.9552022    6.0893338  -12.4996617   -
0.2504738
##        CHits        CRuns         CRBI       CWalks    DivisionW
PutOuts
##    0.3954684    1.4147636    1.0607231   -0.7225505 -123.4055931
0.1923965
##      Assists       Errors   NewLeagueN
##    0.5131110   -7.0115407   70.8774554

predictors = c(names(c_adjr2)[-c(1,11,15)],"Division","NewLeague")
predictors

##  [1] "AtBat"     "Hits"      "Walks"     "Years"     "CAtBat"    "CHits"
##  [7] "CRuns"     "CRBI"      "CWalks"    "PutOuts"   "Assists"   "Errors"
## [13] "Division"  "NewLeague"

formula = reformulate(termlabels = predictors, response = "Salary")
formula

## Salary ~ AtBat + Hits + Walks + Years + CAtBat + CHits + CRuns +
##     CRBI + CWalks + PutOuts + Assists + Errors + Division + NewLeague

m1 = lm(formula, train_data)

y_h1 = predict(m1)
y_test1 = predict(m1,newdata = test_data)

train_mse1 = mean((train_data$Salary - y_h1)^2)
train_mse1

## [1] 88287.08

test_mse1 = mean((test_data$Salary - y_test1)^2)
test_mse1

## [1] 134001.7
```

Forward Subset selection according BIC

```
c_bic = coef(reg2,which.min(reg2_sum$bic))
c_bic

##  (Intercept)        Walks       CAtBat        CHits        CRuns
CRBI
##  127.8159745    6.4746441   -0.4127407    1.1133040    1.0341530
0.8520825
```

```
##       CWalks    DivisionW
##   -0.5432536 -127.3877839
```

```
predictors = c(names(c_bic)[-c(1,8)],"Division")
predictors
```

```
## [1] "Walks"    "CAtBat"   "CHits"    "CRuns"    "CRBI"     "CWalks"
"Division"
```

```
formula = reformulate(termlabels = predictors, response = "Salary")
formula
```

```
## Salary ~ Walks + CAtBat + CHits + CRuns + CRBI + CWalks + Division
```

```
m2 = lm(formula, train_data)

y_h2 = predict(m2)
y_test2 = predict(m2,newdata = test_data)

train_mse2 = mean((train_data$Salary - y_h2)^2)
train_mse2
```

```
## [1] 95658.81
```

```
test_mse2 = mean((test_data$Salary - y_test2)^2)
test_mse2
```

```
## [1] 146540.9
```

Forward Subset selection according Mallow's C_p measure:

```
c_cp = coef(reg2,which.min(reg2_sum$cp))
c_cp
```

```
##  (Intercept)        Walks       CAtBat        CHits        CRuns
CRBI
##  110.1015291    5.2753942   -0.4498891    1.0510348    1.3425171
0.9381944
##       CWalks    DivisionW      PutOuts      Assists       Errors
NewLeagueN
##   -0.5426157 -125.5159872    0.1609710    0.4662319   -7.6994411
74.6774479
```

```
predictors = c(names(c_cp)[-c(1,8,12)],"Division","NewLeague")
predictors
```

```
##  [1] "Walks"     "CAtBat"    "CHits"     "CRuns"     "CRBI"      "CWalks"
##  [7] "PutOuts"   "Assists"   "Errors"    "Division"  "NewLeague"
```

```
formula = reformulate(termlabels = predictors, response = "Salary")
formula
```

```
## Salary ~ Walks + CAtBat + CHits + CRuns + CRBI + CWalks + PutOuts +
##     Assists + Errors + Division + NewLeague
```

```
m3 = lm(formula, train_data)

y_h3 = predict(m3)
y_test3 = predict(m3,newdata = test_data)

train_mse3 = mean((train_data$Salary - y_h3)^2)
train_mse3
```

```
## [1] 90888.44
```

```
test_mse3 = mean((test_data$Salary - y_test3)^2)
test_mse3
```

```
## [1] 145923.9
```

Comparison Matrix

```
Measure = c("Adjusted R sq.","BIC","Mallow's Cp")
Train_MSE_2 = c(train_mse1,train_mse2,train_mse3)
Test_MSE_2 = c(test_mse1,test_mse2,test_mse3)

#Comparision
output = data.frame(Measure,Train_MSE_2,Test_MSE_2)
output
```

```
##          Measure Train_MSE_2 Test_MSE_2
## 1 Adjusted R sq.    88287.08   134001.7
## 2            BIC    95658.81   146540.9
## 3    Mallow's Cp    90888.44   145923.9
```

BACKWARD STEPWISE SELECTION

```
reg3 = regsubsets(Salary ~ ., data = train_data, nvmax = 19, method =
"backward")
reg3_sum = summary(reg3)
```

Backward Subset selection according Aadjusted R^2.

```
c_adjr2 = coef(reg3,which.max(reg3_sum$adjr2))
c_adjr2
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat
CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984
1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts
Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004
0.5618678
```

```
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```
predictors = c(names(c_adjr2)[-c(1,10,14)],"Division","NewLeague")
predictors
```

```
##  [1] "AtBat"     "Hits"      "Walks"     "CAtBat"    "CHits"     "CHmRun"
##  [7] "CRuns"     "CWalks"    "PutOuts"   "Assists"   "Errors"
"Division"
## [13] "NewLeague"
```

```
formula = reformulate(termlabels = predictors, response = "Salary")
formula
```

```
## Salary ~ AtBat + Hits + Walks + CAtBat + CHits + CHmRun + CRuns +
##     CWalks + PutOuts + Assists + Errors + Division + NewLeague
```

```
m1 = lm(formula, train_data)

y_h1 = predict(m1)
y_test1 = predict(m1,newdata = test_data)

train_mse1 = mean((train_data$Salary - y_h1)^2)
train_mse1
```

```
## [1] 88865.02
```

```
test_mse1 = mean((test_data$Salary - y_test1)^2)
test_mse1
```

```
## [1] 134736.7
```

Backward Subset selection according BIC

```
c_bic = coef(reg3,which.min(reg3_sum$bic))
c_bic
```

```
##  (Intercept)        Walks       CAtBat        CHits       CHmRun
DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012
```

```
predictors = c(names(c_bic)[-c(1,6)],"Division")
predictors
```

```
## [1] "Walks"    "CAtBat"   "CHits"    "CHmRun"   "Division"
```

```
formula = reformulate(termlabels = predictors, response = "Salary")
formula
```

```
## Salary ~ Walks + CAtBat + CHits + CHmRun + Division
```

```
m2 = lm(formula, train_data)

y_h2 = predict(m2)
y_test2 = predict(m2,newdata = test_data)

train_mse2 = mean((train_data$Salary - y_h2)^2)
train_mse2

## [1] 97026.43

test_mse2 = mean((test_data$Salary - y_test2)^2)
test_mse2

## [1] 153090.5
```

Backward Subset selection according Mallow's C_p measure:

```
c_cp = coef(reg3,which.min(reg3_sum$cp))
c_cp

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984
1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts
Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004
0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203

predictors = c(names(c_cp)[-c(1,10,14)],"Division","NewLeague")
predictors

##  [1] "AtBat"     "Hits"      "Walks"     "CAtBat"    "CHits"     "CHmRun"
##  [7] "CRuns"     "CWalks"    "PutOuts"   "Assists"   "Errors"
"Division"
## [13] "NewLeague"

formula = reformulate(termlabels = predictors, response = "Salary")
formula

## Salary ~ AtBat + Hits + Walks + CAtBat + CHits + CHmRun + CRuns +
##     CWalks + PutOuts + Assists + Errors + Division + NewLeague

m3 = lm(formula, train_data)

y_h3 = predict(m3)
y_test3 = predict(m3,newdata = test_data)

train_mse3 = mean((train_data$Salary - y_h3)^2)
train_mse3
```

```
## [1] 88865.02

test_mse3 = mean((test_data$Salary - y_test3)^2)
test_mse3

## [1] 134736.7
```

Comparison Matrix

```
Measure = c("Adjusted R sq.","BIC","Mallow's Cp")
Train_MSE_3 = c(train_mse1,train_mse2,train_mse3)
Test_MSE_3 = c(test_mse1,test_mse2,test_mse3)

#Comparision
output = data.frame(Measure,Train_MSE_3,Test_MSE_3)
output

##          Measure Train_MSE_3 Test_MSE_3
## 1 Adjusted R sq.    88865.02   134736.7
## 2            BIC    97026.43   153090.5
## 3    Mallow's Cp    88865.02   134736.7
```

7. Compare the results of the three methods in terms of choice of best model, train MSE amd test MSE.

```
Method = rep(c("Best","Forward","Backward"),each= 3)
Measure_used = rep(Measure,times = 3)
Train_MSE = c(Train_MSE_1,Train_MSE_2,Train_MSE_3)
Test_MSE = c(Test_MSE_1,Test_MSE_2,Test_MSE_3)
compare_mse = data.frame(Method,Measure_used,Train_MSE,Test_MSE)
compare_mse

##     Method   Measure_used Train_MSE Test_MSE
## 1     Best Adjusted R sq.  88935.80 130066.9
## 2     Best            BIC  97026.43 153090.5
## 3     Best    Mallow's Cp  89790.63 125652.3
## 4  Forward Adjusted R sq.  88287.08 134001.7
## 5  Forward            BIC  95658.81 146540.9
## 6  Forward    Mallow's Cp  90888.44 145923.9
## 7 Backward Adjusted R sq.  88865.02 134736.7
## 8 Backward            BIC  97026.43 153090.5
## 9 Backward    Mallow's Cp  88865.02 134736.7
```

Comparison of Training MSE:

- We observe that Adjusted R^2 gives the least training MSE in all three methods, of which forward subset method has the least. Thus we can say the forward method fits the data better than the otherss.

Comparison of Test MSE

- We observe that Mallow's C_p measure performs better than the other measures on an average, of which best Subset method has the least test MSE.

Interpretation

- Forward selection has the lowest training MSE, its test MSE is higher which indicates slight overfitting.
- Backward selection performs similarly to forward but slightly worse.
- Best subset selection, especially using C_p, provides the best balance between bias and variance.

Conclusion

-Evaluating all possible models (best subset) can lead to better generalization. -Stepwise methods (forward/backward) are faster but may not always give the optimal model, as some useful combinations might be left out.

Problem Set 7: Shrinkage Methods

We would be working with the Hitters dataset available in the ISLR library. We have to predict the Salary based on all available variables. First, we make sure that we delete all rows with one or more missing values.

1. We split the data set into a training set and a test set randomly, keeping 100 observations in the test set and the remaining in the training set.

```
set.seed(123)
rm(list=ls())
library(glmnet)

## Warning: package 'glmnet' was built under R version 4.5.3

library(ISLR)

Hitters = na.omit(Hitters)

test_index = sample(1:nrow(Hitters), 100)

test_data = Hitters[test_index, ]
train_data = Hitters[-test_index, ]
```

2. We train the model using least squares and report the train MSE and test MSE.

```
lm_model = lm(Salary ~ ., data = train_data)
summary(lm_model)

##
## Call:
## lm(formula = Salary ~ ., data = train_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -809.76 -182.13  -54.69  138.36 1837.14
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  212.67449  129.87001   1.638 0.103705
## AtBat         -2.04895    0.87954  -2.330 0.021229 *
## Hits           5.85495    3.15579   1.855 0.065612 .
## HmRun          5.43622    9.22155   0.590 0.556448
## Runs          -2.49915    4.03131  -0.620 0.536288
## RBI            2.32756    3.84573   0.605 0.545984
## Walks          6.40986    2.41415   2.655 0.008827 **
## Years         -8.30051   17.14327  -0.484 0.628996
## CAtBat         0.01569    0.19354   0.081 0.935512
## CHits         -0.54632    1.02039  -0.535 0.593204
## CHmRun        -2.38189    2.49628  -0.954 0.341604
## CRuns          1.56320    1.05024   1.488 0.138840
## CRBI           1.26993    1.02979   1.233 0.219524
## CWalks        -0.95946    0.45825  -2.094 0.038049 *
## LeagueN       44.32184   98.13766   0.452 0.652220
## DivisionW   -149.54532   56.87996  -2.629 0.009496 **
## PutOuts        0.41589    0.11170   3.723 0.000282 ***
## Assists        0.43589    0.28787   1.514 0.132176
## Errors        -5.50162    5.83866  -0.942 0.347642
## NewLeagueN   -15.79562   97.46148  -0.162 0.871479
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 346.5 on 143 degrees of freedom
## Multiple R-squared:  0.4782, Adjusted R-squared:  0.4088
## F-statistic: 6.897 on 19 and 143 DF,  p-value: 1.144e-12
```

```r
y_hat = predict(lm_model, newdata = train_data)
y_test = predict(lm_model, newdata = test_data)

train_mse = mean((train_data$Salary-y_hat)^2)
train_mse
```

```
## [1] 105334.2
```

```r
test_mse = mean((test_data$Salary-y_test)^2)
test_mse
```

```
## [1] 87803.91
```

3.We fit ridge regression models on the training set, with grid values of $\lamda$. We report the following:

(i) the dimension of the coefficient matrix so obtained.

(ii) The l_2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. We comment on the output.

(iii) Test MSE corresponding to these two choices of λ.

```r
lambda_grid = 10^seq(10, -2, length = 100)
x_train = model.matrix(Salary~.,train_data)[,-1]
x_test = model.matrix(Salary~.,test_data)[,-1]
head(x_train)
```

```
##                   AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun
## -Alan Ashby         315   81     7   24  38    39    14   3449   835     69
## -Alvin Davis        479  130    18   66  72    76     3   1624   457     63
## -Andre Dawson       496  141    20   65  78    37    11   5628  1575    225
## -Alfredo Griffin    594  169     4   74  51    35    11   4408  1133     19
## -Andres Thomas      323   81     6   26  32     8     2    341    86      6
## -Andre Thornton     401   92    17   49  66    65    13   5206  1332    253
##                   CRuns CRBI CWalks LeagueN DivisionW PutOuts Assists
Errors
## -Alan Ashby         321  414    375       1         1     632      43
10
## -Alvin Davis        224  266    263       0         1     880      82
14
## -Andre Dawson       828  838    354       1         0     200      11
3
## -Alfredo Griffin    501  336    194       0         1     282     421
25
## -Andres Thomas       32   34      8       1         1     143     290
19
## -Andre Thornton     784  890    866       0         0       0       0
0
##                   NewLeagueN
## -Alan Ashby                1
## -Alvin Davis               0
## -Andre Dawson              1
## -Alfredo Griffin           0
## -Andres Thomas             1
## -Andre Thornton            0
```

```r
ridge_model=glmnet(x_train, train_data$Salary, alpha = 0,lambda=lambda_grid)
dim(coef(ridge_model))
```

```
## [1]  20 100
```

(i) Dimension of Coefficient Matrix

```r
dim(coef(ridge_model))
```

```
## [1]  20 100
```

(ii) Coefficients for 50th and 60th value of λ respectively

```r
# 50th value
coef_50 = coef(ridge_model)[, 50]
coef_50
```

```
##   (Intercept)         AtBat          Hits         HmRun          Runs
## 438.509969440   0.029684297   0.110523230   0.560867468   0.178502584
```

```
##           RBI         Walks         Years        CAtBat         CHits
##   0.252163513   0.238337160   1.002466622   0.002796464   0.010194612
##        CHmRun         CRuns          CRBI        CWalks       LeagueN
##   0.077846378   0.020089803   0.021957346   0.021866228   0.432252998
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN
##  -6.916280599   0.018294166   0.002998934  -0.082552703   0.741567242
```

```r
cat("\n")

#60th Value
coef_60 = coef(ridge_model)[, 60]
coef_60
```

```
##  (Intercept)        AtBat         Hits        HmRun         Runs
RBI
## 105.63996577   0.08310078   0.51246876   2.09635144   0.60993278
1.14623958
##        Walks        Years       CAtBat        CHits       CHmRun
CRuns
##   1.01677160   2.77366963   0.01043635   0.04276159   0.27032634
0.08012067
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts
Assists
##   0.09127173   0.05840151  18.35350369 -62.46410535   0.14784491
0.04002532
##       Errors   NewLeagueN
##  -0.86723778  12.21693330
```

(ii). L2 Norm of Coefficients

```r
l2_norm = function(beta){
  sqrt(sum(beta^2))
}

# l2 norm of 50th lambda
l2_50 = l2_norm(coef_50)
l2_50
```

```
## [1] 438.5671
```

```r
# l2 norm of 60th lambda
l2_60 = l2_norm(coef_60)
l2_60
```

```
## [1] 124.7542
```

(iii). Test MSE for 50th and 60th λ

```r
# Predictions
ridge_pred_50 = predict(ridge_model, s = lambda_grid[50], newx = x_test)
ridge_pred_60 = predict(ridge_model, s = lambda_grid[60], newx = x_test)
```

```r
# Test MSE for 50th lambda
Test_mse_50 = mean((test_data$Salary - ridge_pred_50)^2)
Test_mse_50

## [1] 167303.1

# Test MSE for the 60th lambda
Test_mse_60 = mean((test_data$Salary - ridge_pred_60)^2)
Test_mse_60

## [1] 97935.56
```

We observe that the test MSE is dropping as Lamda increases.